Streaming of Big Data

in Spark

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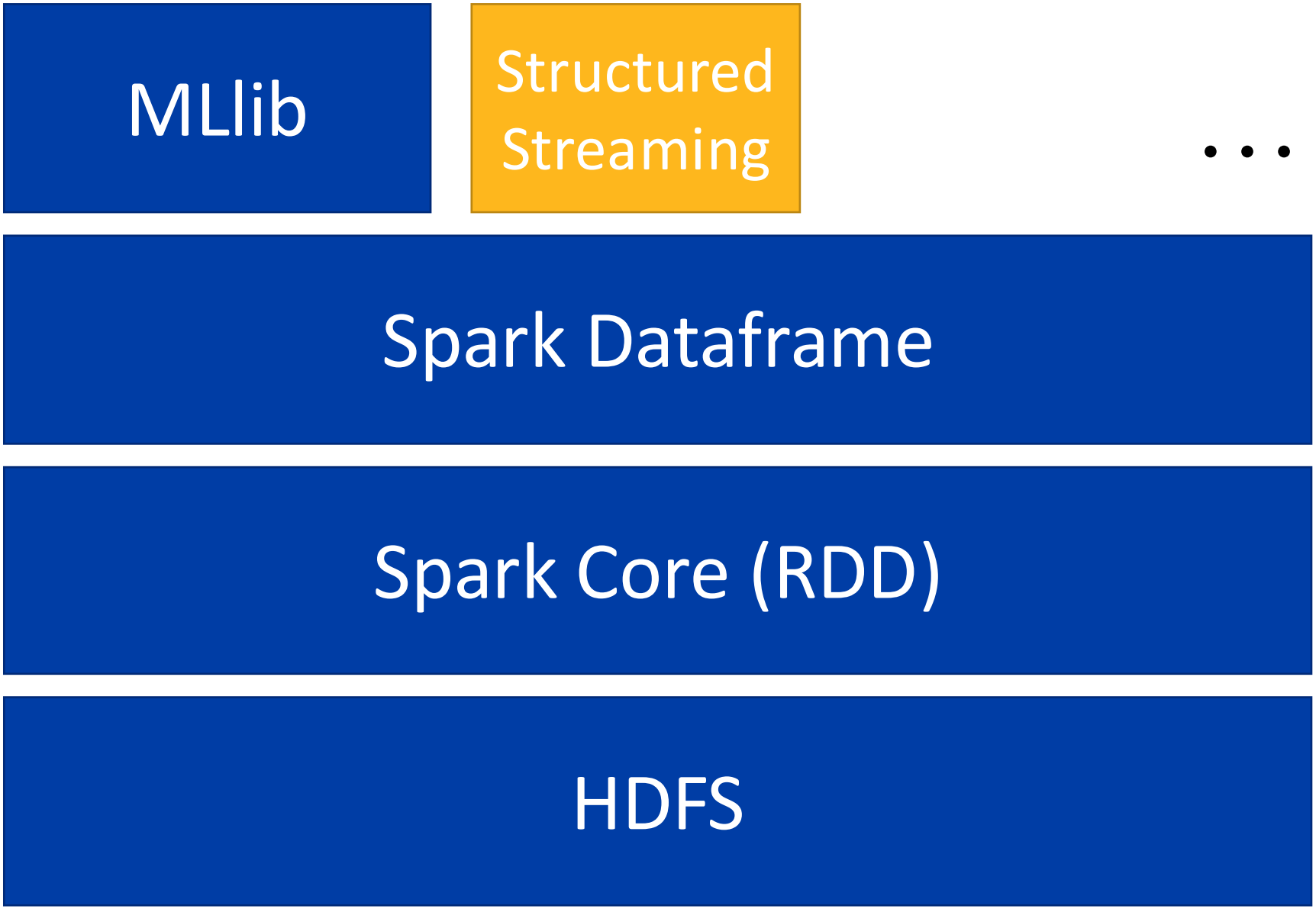
Structured Streaming



Streaming in Spark

- Spark Streaming
  - Extends the core RDD API to support streaming
  - Not recommended
- Structured Streaming
  - Extends the Dataframe API (SparkSQL)
  - More efficient and recommended

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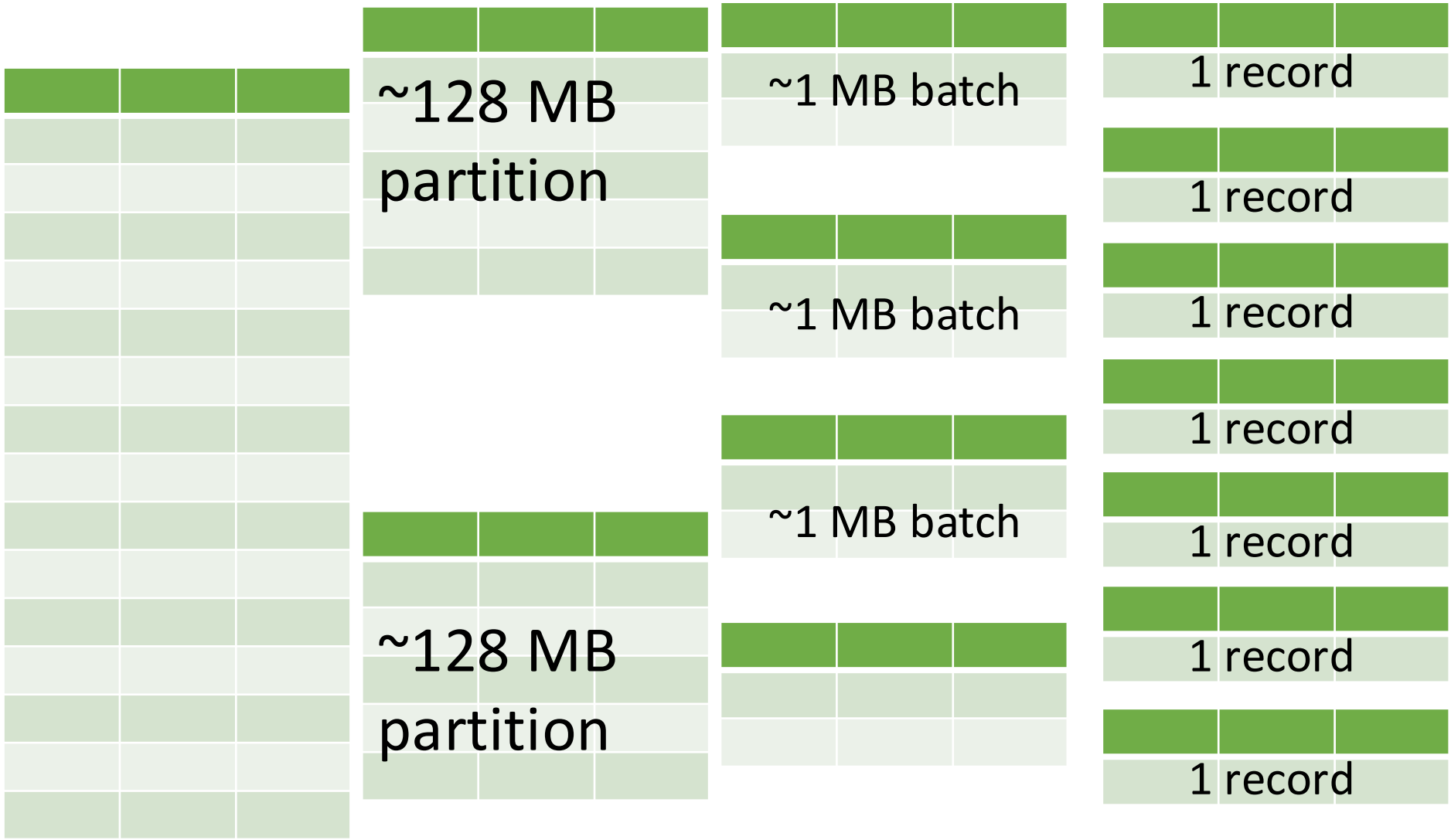
Types of Streaming



Input Dataframe

Batch processing (SparkSQL)

Micro batching (StructuredStreaming)

Continuous streaming

Word Count Example

```
val spark = SparkSession
  .builder
  .appName("StructuredNetworkWordCount")
  .getOrCreate()

// Create DataFrame from a network connection
val lines = spark.readStream
  .format("socket")
  .option("host", "localhost")
  .option("port", 9999)
  .load()

// Split the lines into words
val words = lines.as[String].flatMap(_.split(" "))

// Generate running word count
val wordCounts = words.groupBy("value").count()

// Start running the query that prints the running counts to the console
val query = wordCounts.writeStream
  .outputMode("complete")
  .format("console")
  .start()

query.awaitTermination()
```

Input/Output

```
apache spark
apache hadoop
```

```
-------------------------------------------
Batch: 0
-------------------------------------------
+------+-----+
| value|count|
+------+-----+
|apache|    1|
| spark|    1|
+------+-----+

-------------------------------------------
Batch: 1
-------------------------------------------
+------+-----+
| value|count|
+------+-----+
|apache|    2|
| spark|    1|
|hadoop|    1|
+------+-----+
...
```

Output Modes

- Complete
  - Produce the entire output after each batch
  - Usually useful for debugging and for small output
- Append
  - Only produce new output records
  - No changes are allowed
- Update
  - Produce both new records and modified records

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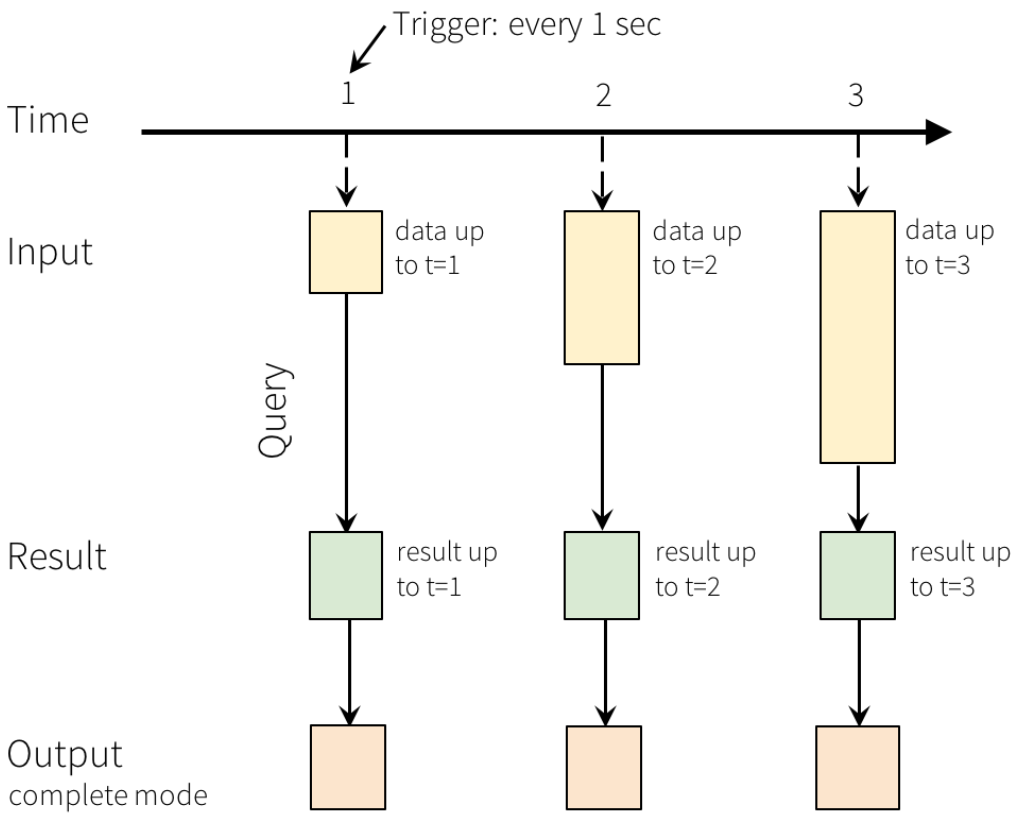
Complete Output Mode



Programming Model for Structured Streaming

Figure: http://spark.apache.org/docs/latest/structured-streaming-programming-guide.html

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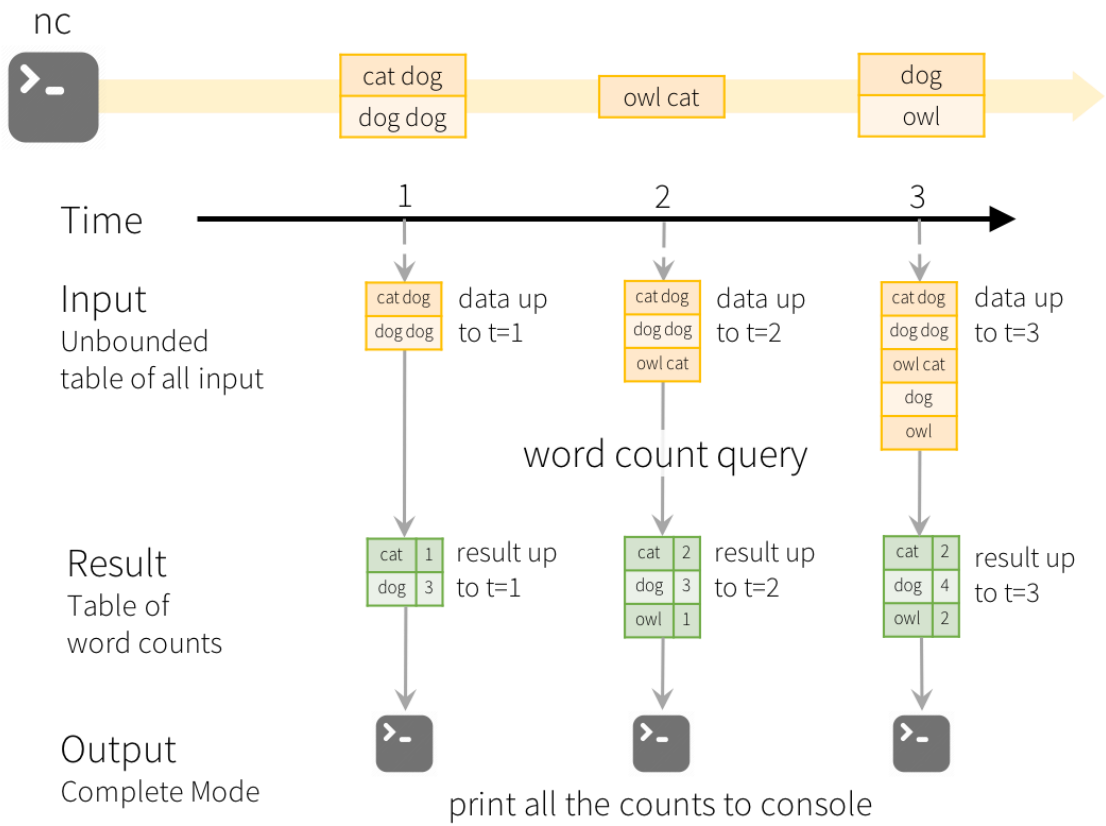
Complete Output Mode



Model of the Quick Example

Figure: http://spark.apache.org/docs/latest/structured-streaming-programming-guide.html

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Timestamps

- In some cases, data might come with its own timestamps
  - Weather data
  - Crimes/Incidents
  - Moving objects
  - Tweets
- Structured Streaming allows developers to utilize these timestamps
- The assumption is that the data is already (partially) sorted by this timestamp

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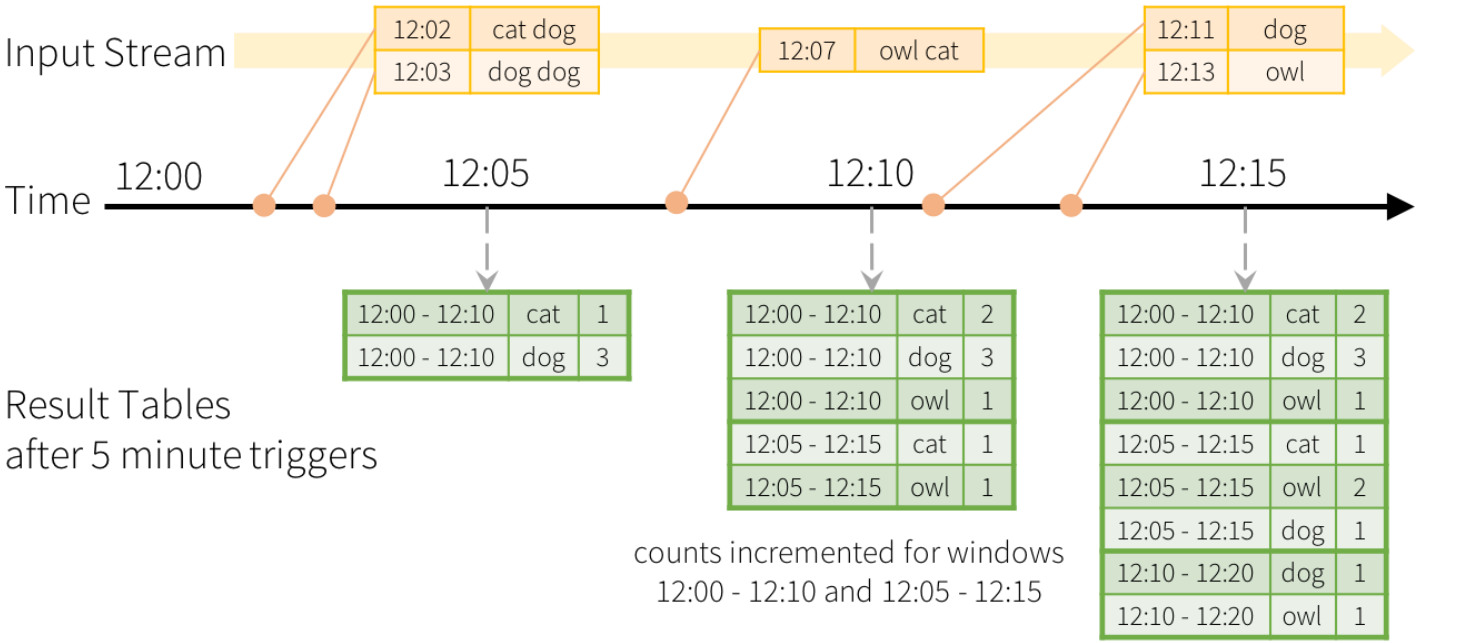
Window Operations

```
// Group the data by window and word and compute the
count of each group
val windowedCounts = words.groupBy(
  window($"timestamp", "10 minutes", "5 minutes"),
  $"word"
).count()
```

Windowed Processing



Windowed Grouped Aggregation
with 10 min windows, sliding every 5 mins

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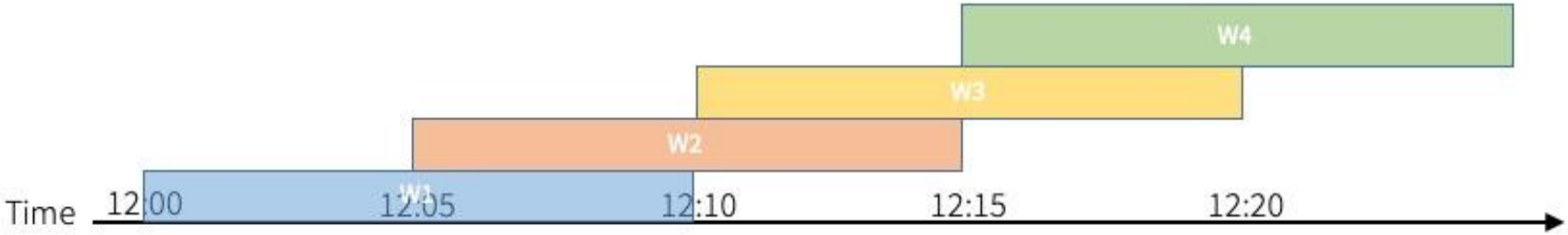
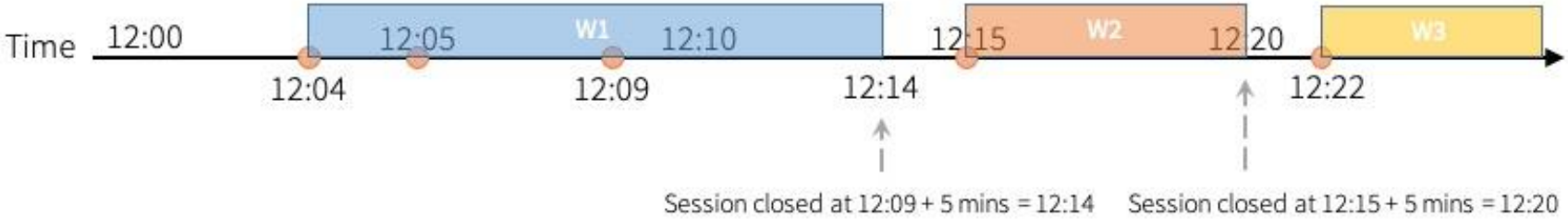
Types of Windows

Tumbling Windows (5 mins)

Sliding Windows (10 mins, slide 5 mins)



Session Windows (gap duration 5 mins)



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Watermarking

- The input data might not be totally ordered
- For example, when merging multiple ordered streams
- Why not ensuring a total ordering?
- Watermarking is a Structured Streaming feature for informing Spark how much error to tolerate

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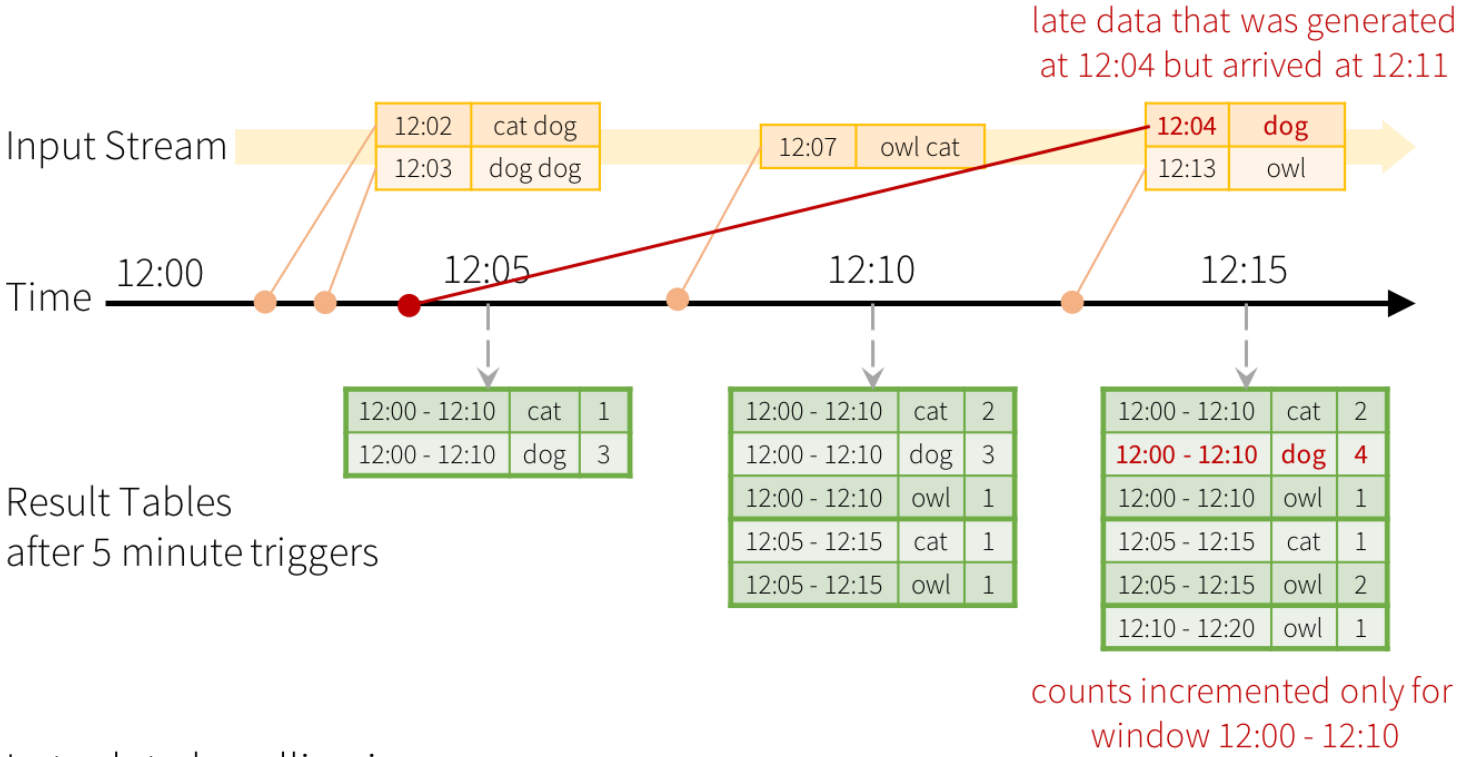
Watermarking



Late data handling in
Windowed Grouped Aggregation

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Watermarking

```
// Group the data by session window and userId, and compute the
count of each group
val sessionizedCounts = events
    .withWatermark("timestamp", "10 minutes")
    .groupBy(
        session_window($"timestamp", "5 minutes"),
        $"userId")
    .count()
```

Session Windows

```
// Group the data by session window and userId, and compute the
count of each group
val sessionizedCounts = events
    .withWatermark("timestamp", "10 minutes")
    .groupBy(
        session_window($"timestamp", "5 minutes"),
        $"userId")
    .count()
```

Session Windows

```
// Session window with dynamic gap
val sessionWindow = session_window($"timestamp",
     when($"userId" === "user1", "5 seconds")
      .when($"userId" === "user2", "20 seconds")
      .otherwise("5 minutes"))

// Group the data by session window and userId, and compute the
count of each group
val sessionizedCounts = events
    .withWatermark("timestamp", "10 minutes")
    .groupBy(
        Column(sessionWindow),
        $"userId")
    .count()
```

Further Reading

- Official Programming Guide
  - http://spark.apache.org/docs/latest/structured-streaming-programming-guide.html